Adaptive-OCGCN: Ambiguity-Aware Overlap Assignment for Scalable Graph Neural Networks

Mahmood Amintoosi[1,2*]

[1]Computer Science Division, Department of Applied Mathematics, Faculty of Mathematical Sciences, Ferdowsi University of Mashhad, P. O. Box 1159, Mashhad 91775, Iran.

[2]Graph Theory and its Applications Lab., Faculty of Mathematical Sciences, Ferdowsi University of Mashhad, Mashhad, Iran.

Corresponding author(s). E-mail(s): m.amintoosi@um.ac.ir;

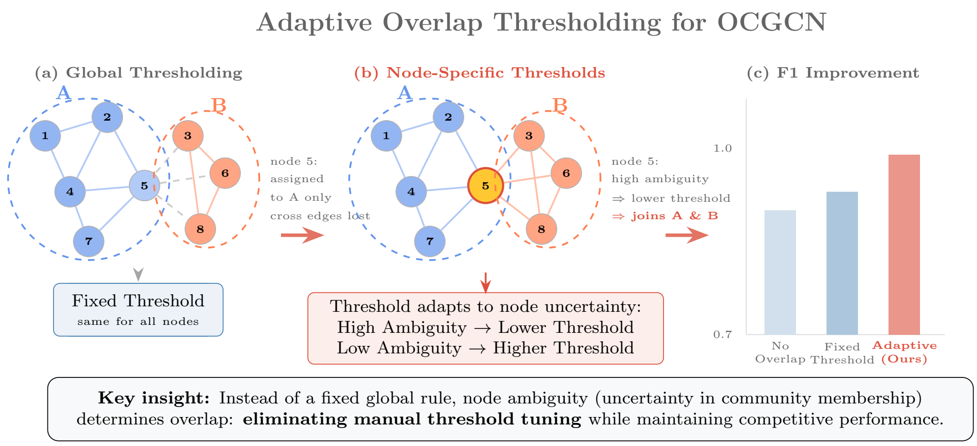
Abstract

Graph Convolutional Networks (GCNs) have demonstrated strong performance on graph learning tasks, but their scalability remains challenging for large-scale graphs. Cluster-based training approaches, such as Cluster-GCN, address this issue by partitioning large graphs into smaller subgraphs; however, hard partitioning may discard important inter-cluster information. Overlapping Cluster-GCN (OCGCN) alleviates this limitation by allowing boundary nodes to participate in multiple clusters through overlap assignment. Nevertheless, OCGCN relies on a fixed global Winner Membership Closeness (WMC) threshold, which requires dataset-specific tuning and does not consider the uncertainty of individual node memberships.

In this work, we propose Adaptive-OCGCN, an ambiguity-aware overlap assignment framework that replaces global thresholding with node-specific adaptive decisions. The proposed approach exploits community-membership ambiguity to identify nodes whose assignments are inherently uncertain. Specifically, we introduce three adaptive strategies: Entropy-Adaptive WMC, which utilizes normalized membership entropy, Margin-Adaptive WMC, which measures ambiguity through the gap between the two most likely community memberships, and Hybrid-Adaptive WMC, which combines both signals.

We evaluate the proposed methods on six benchmark graph datasets (Cora, CiteSeer, PubMed, ACM, DBLP, and IMDB) using multiple random seeds and compare them with fixed-threshold OCGCN baselines. The results demonstrate that adaptive overlap assignment matches tuned fixed-threshold OCGCN while removing the need for dataset-specific WMC grid search. Across 20 seeds and six datasets, Margin-Adaptive WMC outperforms the untuned default threshold on four of six datasets, with a significant pooled paired gain, and adaptive methods also improve performance on four of six datasets at comparable overlap ratios. Furthermore, ambiguity analysis shows that high-entropy nodes contain 41–95% overlap candidates, whereas low-entropy nodes contain only 0–17%, confirming that membership uncertainty provides a meaningful criterion for overlap selection. Overall, Adaptive-OCGCN provides a practical and principled approach for scalable graph neural network training by integrating uncertainty-aware overlap assignment into partition-based learning.

Keywords: Graph Convolutional Networks, Overlapping Clustering, Cluster-GCN, Membership Ambiguity, Adaptive Thresholding



Contents

1 Introduction

Graph Convolutional Networks (GCNs) have become a foundational tool for semi-supervised node classification on graph-structured data (Kipf and Welling 2017). However, applying GCNs to large-scale graphs remains challenging due to neighborhood explosion, where each convolutional layer requires aggregating information from increasingly large neighborhoods, resulting in substantial memory and computational costs (Chiang et al. 2019). To address this issue, Cluster-GCN partitions a large graph into smaller subgraphs and performs mini-batch training on these clusters, achieving improved scalability while preserving most local graph structures (Chiang et al. 2019). The underlying assumption is that nodes within the same partition contain sufficient structural information and that the loss of cross-cluster dependencies is limited.

Nevertheless, real-world graphs often exhibit overlapping community structures, where individual nodes naturally participate in multiple semantic or structural groups. For example, papers in citation networks may connect multiple research areas, while entities in heterogeneous networks may simultaneously play different roles. When traditional partitioning assigns such boundary nodes exclusively to a single cluster, important cross-community information may be discarded, leading to incomplete node representations.

Overlapping Cluster-GCN (OCGCN), introduced in our earlier study (Amintoosi 2021), was an initial attempt to incorporate overlapping community membership into partition-based GCN training. Specifically, OCGCN employs a non-negative matrix factorization based community detection method (DANMF) to obtain soft community memberships and allows nodes with significant membership in multiple clusters to participate in multiple cluster subgraphs. This overlap mechanism enables the GCN model to access richer structural contexts during training.

Despite its effectiveness, OCGCN determines overlap membership using a single global Winner Membership Closeness (WMC) threshold. Such a global decision rule assumes that all nodes exhibit similar levels of community certainty, which is rarely true in real-world networks. A node with nearly uniform membership across several communities represents a highly ambiguous case, whereas a node with one dominant community and a weak secondary membership has a substantially different structural meaning. However, both nodes are treated identically under a fixed global threshold. Furthermore, the appropriate WMC value varies across datasets, requiring additional hyperparameter tuning and limiting the practicality of overlap-aware training.

In this work, we introduce *Adaptive-OCGCN*, an ambiguity-aware overlap assignment framework that replaces the fixed global overlap criterion with node-specific adaptive decisions. The central idea is that community-membership ambiguity provides a natural signal for identifying nodes that benefit most from multi-cluster participation. We propose three complementary adaptive strategies: *Entropy-Adaptive WMC*, which uses normalized membership entropy to identify nodes with distributed community affiliations, *Margin-Adaptive WMC*, which exploits the gap between the two largest membership probabilities to detect nodes located near community boundaries, and *Hybrid-Adaptive WMC*, which combines both signals. All strategies preserve the original OCGCN formulation while introducing node-level adaptivity through an ambiguity-aware adjustment mechanism.

Unlike previous overlap-aware approaches that treat overlap selection as a global partitioning decision, Adaptive-OCGCN formulates overlap assignment as a node-level uncertainty-aware learning problem. The main contributions of this work are summarized as follows:

1. We identify a fundamental limitation of existing overlap-aware Cluster-GCN training: fixed global overlap criteria ignore node-level variation in community-membership ambiguity.
2. We propose *Adaptive-OCGCN*, an ambiguity-aware overlap assignment framework with three adaptive strategies, Entropy-Adaptive WMC, Margin-Adaptive WMC, and their Hybrid combination, which derive node-specific overlap decisions from membership uncertainty.
3. We demonstrate through extensive experiments on six benchmark graph datasets that adaptive overlap assignment achieves competitive or improved performance compared with fixed-threshold OCGCN while substantially reducing dataset-specific WMC tuning.
4. We perform a detailed ambiguity analysis showing that membership uncertainty strongly correlates with overlap suitability, providing empirical evidence that ambiguity is a principled criterion for selecting nodes for overlapping training.

2 Related Work

This section surveys three lines of research relevant to our work: scalable graph neural network training (Section 2.5), overlapping community detection and soft membership modeling (Section 2.7), and overlap-aware partition-based training (Section 2.6). We also review recent advances in adaptive and uncertainty-aware graph learning (Section 2.8) and motivate our ambiguity-based approach.

2.1 Scalable Graph Neural Network Training

Graph Convolutional Networks (GCNs) have become a fundamental approach for representation learning on graph-structured data by iteratively aggregating information from neighboring nodes (Kipf and Welling 2017). However, applying GCNs to large-scale graphs remains challenging due to the neighborhood explosion problem, where multi-layer message passing requires accessing exponentially increasing neighborhoods.

Several approaches have been proposed to improve the scalability of GNN training. GraphSAGE (Hamilton et al. 2017) and FastGCN (Chen et al. 2018) reduce computational costs through neighborhood sampling, while VR-GCN (Dai et al. 2018) introduces variance reduction techniques by reusing historical node representations. Although sampling-based approaches improve efficiency, they may introduce approximation errors and remain challenging for very deep architectures.

Cluster-GCN (Chiang et al. 2019) follows a different strategy by partitioning a large graph into smaller clusters and performing mini-batch training on cluster-induced subgraphs. By restricting message passing within each partition, Cluster-GCN significantly reduces memory and computational requirements while maintaining competitive performance. However, this approach relies on a hard partitioning assumption: nodes are assigned to a single cluster, and cross-cluster interactions are largely ignored. Consequently, nodes located near community boundaries may lose important structural information during training.

2.2 Overlapping Community Learning and Soft Membership Modeling

Many real-world networks exhibit overlapping community structures, where individual nodes simultaneously participate in multiple semantic or structural groups. Unlike traditional hard clustering methods, overlapping community detection approaches aim to learn soft membership representations that capture the degree of association between nodes and communities.

Existing overlapping community detection methods include clique-based approaches, label propagation techniques, and matrix factorization based methods (Shang et al. 2014). Non-negative matrix factorization (NMF) methods represent each node using a membership vector over latent communities. Deep Adversarial Non-negative Matrix Factorization (DANMF) (Tran et al. 2018) extends this idea by combining deep autoencoding and adversarial learning to obtain informative community representations.

Recent studies have further explored graph neural network based approaches for overlapping community discovery. He et al. (He 2022) proposed a semi-supervised overlapping community detection framework using graph convolutional autoencoders. Yuan et al. (Yuan 2022) investigated overlapping community detection using graph convolutional networks on complex networks. UCoDe (Moradan 2023) introduced a unified graph convolutional framework for community detection, while Li et al. (Li 2023) proposed a dual graph neural network architecture for overlapping community detection. Sismanis et al. (Sismanis 2024) further explored graph attention networks for identifying overlapping communities.

Although these approaches effectively model overlapping community structures, their primary objective is community discovery. They do not consider how overlapping memberships can be exploited to improve scalable graph neural network training under partition-based learning frameworks.

2.3 Overlap-Aware Partition-Based GNN Training

While most previous studies have investigated overlapping communities for network analysis and community discovery, Overlapping Cluster-GCN (OCGCN) (Amintoosi 2021) introduced a different perspective by utilizing overlapping memberships to improve scalable graph neural network training.

OCGCN extends Cluster-GCN by incorporating soft community memberships obtained from DANMF into the graph partitioning process. Instead of assigning every node to a single cluster, OCGCN allows nodes with significant memberships in multiple communities to participate in multiple cluster subgraphs. This strategy alleviates the information loss caused by hard partitioning and provides richer structural contexts for GCN training.

The overlap assignment in the original OCGCN framework is controlled by a Winner Membership Closeness (WMC) threshold. A node is assigned to cluster c if its membership value satisfies:

$$P(i, c) \geq \mathrm{WMC} \cdot \max_{c'} P(i, c').$$

Although this mechanism improves boundary-node representation, it applies the same global threshold to all nodes. Such a criterion assumes that all nodes have similar levels of community certainty, while real-world graphs contain both highly confident nodes and highly ambiguous nodes with competing community memberships. Therefore, overlap selection in OCGCN remains dependent on dataset-specific threshold tuning rather than node-level structural characteristics.

2.4 Adaptive and Uncertainty-Aware Graph Learning

Adaptive mechanisms have recently attracted increasing attention in graph representation learning. Graph Attention Networks (GAT) (Veličković et al. 2018), for example, learn node-specific importance coefficients to dynamically control information aggregation. Similarly, adaptive neighborhood selection approaches adjust message-passing structures according to graph topology and learning objectives (Peng 2021; Guang 2024).

Beyond adaptive aggregation, uncertainty-aware learning has emerged as an important direction for understanding model reliability and decision confidence. Gawlikowski et al. (Gawlikowski 2023) provided a comprehensive survey of uncertainty estimation methods in deep neural networks, highlighting the importance of uncertainty modeling for reliable prediction. Several graph learning studies have also incorporated confidence-aware mechanisms for improving robustness and reliability (Kwon 2022; Park 2023).

However, existing adaptive and uncertainty-aware graph learning approaches mainly focus on message aggregation, neighborhood selection, or prediction reliability. They do not address the problem of deciding which nodes should participate in multiple graph partitions during scalable GNN training.

In this work, we bridge this gap by introducing membership ambiguity as a node-level uncertainty signal for overlap assignment. Unlike previous approaches that treat overlap selection as a global partitioning decision, Adaptive-OCGCN formulates overlap assignment as an uncertainty-aware node-level learning problem.

2.5 Cluster-GCN

Cluster-GCN (Chiang et al. 2019) accelerates GCN training by exploiting graph partitioning. Given a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with $N = |\mathcal{V}|$ nodes, Cluster-GCN partitions $\mathcal{V}$ into K disjoint clusters $\{C_1, \ldots, C_K\}$.

Instead of performing full-batch training on the entire graph, a GCN is trained on cluster-induced subgraphs $\mathcal{G}[C_k]$, where message passing is restricted to nodes within each cluster. This reduces memory consumption and computational complexity, making GCN training feasible for large-scale graphs.

The standard GCN layer is defined as:

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}\right), \tag{1}$$

where $\tilde{A} = A + I_N$ represents the adjacency matrix with self-loops, $\tilde{D}$ is the corresponding degree matrix, $W^{(l)}$ is a trainable weight matrix, and σ denotes the activation function.

The main assumption of Cluster-GCN is that most important structural information is preserved within clusters. However, when nodes lie at the intersection of multiple communities, removing cross-cluster connections may lead to incomplete neighborhood information and degraded representations.

2.6 Overlapping Cluster-GCN

Overlapping Cluster-GCN (OCGCN) (Amintoosi 2021) extends Cluster-GCN by relaxing the assumption of disjoint graph partitions. The main idea is to identify boundary nodes with significant memberships in multiple communities and allow them to participate in multiple cluster subgraphs.

The OCGCN framework consists of four main steps:

1. Soft Community Assignment. A community detection algorithm based on DANMF produces a soft membership matrix $P \in \mathbb{R}^{N \times K}$, where $P(i, c)$ indicates the membership strength of node i in cluster c.

2. Overlap Assignment. For each node i, OCGCN selects clusters satisfying:

$$P(i, c) \geq \text{WMC} \times \max_{c'} P(i, c'). \tag{2}$$

where $\text{WMC} \in (0, 1]$ is the Winner Membership Closeness threshold.
3. Overlapping Subgraph Construction. Nodes selected for multiple clusters are duplicated across corresponding cluster subgraphs while preserving their original features and labels.
4. Training and Prediction. Independent GCN models are trained on cluster subgraphs. For overlapping nodes, predictions from multiple assigned clusters are aggregated.

By allowing boundary nodes to appear in multiple partitions, OCGCN reduces the information loss caused by hard graph partitioning. However, the original framework uses a single global WMC threshold for all nodes. This ignores the fact that membership distributions may have different levels of certainty, motivating an adaptive overlap assignment strategy.

2.7 Soft Community Membership and DANMF

DANMF (Tran et al. 2018) is a deep adversarial non-negative matrix factorization method that learns latent community representations. Unlike hard clustering, DANMF provides soft membership vectors, allowing each node to have different association strengths with multiple communities.

The resulting normalized membership matrix is denoted by $P \in \mathbb{R}^{N \times K}$, where:

$$P(i, c) \geq 0, \qquad \sum_{c=1}^{K} P(i, c) = 1. \tag{3}$$

These membership distributions provide the foundation for measuring node-level community ambiguity in our adaptive overlap assignment framework.

2.8 Community Membership Ambiguity

The central hypothesis of this work is that nodes with uncertain community memberships are more suitable candidates for overlap assignment. To quantify this uncertainty, we consider two complementary measures derived from the membership distribution $P(i, :)$.

Normalized Entropy.

Entropy measures the global uncertainty of the membership distribution:

$$H_{\text{norm}}(i) = \frac{-\sum_{c=1}^{K} P(i, c) \log P(i, c)}{\log K}. \tag{4}$$

A value close to zero indicates a concentrated membership distribution, where one community dominates. A value close to one indicates that the node has similar memberships across communities.

Membership Margin Ambiguity.
Entropy captures uncertainty across all communities, but it does not specifically focus on competition between the most likely communities. Therefore, we additionally define a margin-based ambiguity score:

$$A(i) = 1 - (p_1 - p_2), \tag{5}$$

where p_1 and p_2 denote the largest and second-largest membership values of node i.

A small margin between the two dominant communities indicates that the node lies near a community boundary. Thus, high values of $A(i)$ correspond to nodes with ambiguous cluster assignments.

Entropy and margin ambiguity capture different aspects of uncertainty. Entropy measures the overall distribution spread, whereas margin ambiguity focuses on the competition between the two strongest communities. These complementary characteristics motivate the two adaptive overlap assignment strategies proposed in this work.

3 Proposed Method

Given a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with node features $X \in \mathbb{R}^{N \times d}$ and labels Y for a subset of nodes, the objective of semi-supervised node classification is to learn a graph neural network model that predicts labels for unlabeled nodes. For large-scale graphs, Cluster-GCN addresses the computational challenge by partitioning the graph into K disjoint subgraphs. However, hard partitioning may remove important cross-community interactions, especially for nodes located near community boundaries. OCGCN alleviates this limitation by introducing an overlapping partitioning mechanism. Given a soft community membership matrix

$$P \in \mathbb{R}^{N \times K}, \tag{6}$$

where $P(i, c)$ denotes the membership strength of node i in community c, OCGCN determines the cluster assignments of each node using a global Winner Membership Closeness threshold:

$$\mathcal{C}_i = \left\{ c \mid P(i, c) \geq \mathrm{WMC} \times \max_{c'} P(i, c') \right\}. \tag{7}$$

Although effective, this formulation applies the same WMC value to all nodes, which does not consider the confidence of individual community memberships. For example, a node with an ambiguous membership distribution

$$P(i, :) = [0.35, 0.33, 0.32]$$

and a node with a confident assignment

$$P(j, :) = [0.90, 0.08, 0.02]$$

are processed using the same overlap criterion, despite having fundamentally different structural characteristics. The problem addressed in this work is therefore to determine node-specific overlap assignments by exploiting membership ambiguity, while preserving the scalability advantages of partition-based GNN training. More formally, we seek an adaptive overlap assignment function:

$$\mathcal{C}_i = f(P(i,:), \lambda), \tag{8}$$

where $f(\cdot)$ determines the cluster participation of node i based on its membership distribution, and λ controls the strength of adaptation. The desired adaptive assignment mechanism should satisfy three properties: ambiguity awareness (nodes with uncertain community memberships should have higher probability of being assigned to multiple clusters), confidence preservation (nodes with dominant memberships should avoid unnecessary overlap), and hyperparameter efficiency (the method should reduce dependence on dataset-specific threshold tuning).

This section introduces our adaptive overlap assignment framework. Section 3.1 provides a high-level overview of the adaptive threshold mechanism. Sections 3.2–3.4 present three complementary adaptive strategies based on entropy, margin, and hybrid ambiguity signals. Section 3.5 compares these strategies, and Section **??** summarizes the complete procedure.

3.1 Overview

The original OCGCN determines overlap assignments using a single global WMC threshold. However, nodes in a graph exhibit different levels of community-membership certainty. Therefore, we reformulate overlap selection as an ambiguity-aware decision process, where each node is assigned an adaptive threshold according to its membership distribution.

Let $a_i \in [0,1]$ denote the ambiguity score of node i. A high value of a_i indicates that the node has uncertain community membership and is a potential boundary node. The adaptive threshold is defined as:

$$\mathrm{WMC}_i = \max\left(0.01,\ \mathrm{WMC}_{\mathrm{base}}\left(1 - \lambda a_i\right)\right), \tag{9}$$

where $\mathrm{WMC}_{\mathrm{base}}$ is the reference threshold inherited from the original OCGCN formulation and $\lambda \in [0,1]$ controls the adaptation strength.

When $\lambda = 0$, the proposed method reduces to the original OCGCN with a fixed global threshold. Increasing λ allows ambiguous nodes to receive lower thresholds and consequently participate in additional cluster subgraphs. To avoid excessive overlap, WMC_i is bounded below by 0.01. The same lower bound applies to all adaptive variants introduced below.

Using the adaptive threshold, node i is assigned to cluster c if:

$$P(i,c) \geq \mathrm{WMC}_i \max_{c'} P(i,c'). \tag{10}$$

The proposed framework is independent of the community detection algorithm and can be applied whenever a soft membership matrix is available. In this work, we use DANMF-generated memberships as the input representation.

3.2 Entropy-Adaptive WMC

Entropy-Adaptive WMC defines the ambiguity score using the normalized entropy of the membership distribution:

$$a_i = H_{\mathrm{norm}}(i), \tag{11}$$

which results in:

$$\mathrm{WMC}_i = \max\left(0.01,\ \mathrm{WMC}_{\mathrm{base}}(1 - \lambda H_{\mathrm{norm}}(i))\right). \tag{12}$$

Mechanism.

Nodes with concentrated membership distributions have low entropy and retain a threshold close to $\mathrm{WMC}_{\mathrm{base}}$. Consequently, confident nodes are unlikely to be assigned to unnecessary additional clusters. In contrast, nodes with high entropy distribute their membership probability among several communities and receive a lower threshold, allowing them to participate in multiple cluster subgraphs.

Advantages.

Entropy considers the complete membership distribution rather than only the two most probable clusters. Therefore, it is suitable for graphs where nodes may simultaneously relate to multiple communities. Furthermore, normalization makes the ambiguity score comparable across datasets with different numbers of detected communities.

3.3 Margin-Adaptive WMC

Margin-Adaptive WMC uses the ambiguity derived from the difference between the two largest membership probabilities:

$$a_i = A(i) = 1 - (p_1 - p_2), \tag{13}$$

leading to:

$$\mathrm{WMC}_i = \max\left(0.01,\ \mathrm{WMC}_{\mathrm{base}}(1 - \lambda A(i))\right). \tag{14}$$

Mechanism.

When the largest membership probability is substantially higher than the second largest, the node has a clear community assignment and receives a threshold close to the base value. When the top two memberships are close, the node represents a community boundary and is assigned a lower threshold to preserve cross-community information.

Advantages.

Unlike entropy, margin ambiguity focuses specifically on competition between the two dominant communities. This makes it particularly suitable for graphs where overlapping behavior mainly occurs between pairs of communities rather than across multiple communities.

3.4 Hybrid-Adaptive WMC

Hybrid-Adaptive WMC combines the entropy and margin ambiguity signals into a single score:

$$a_i = \max\left(H_{\text{norm}}(i),\ A(i)\right), \tag{15}$$

leading to:

$$\text{WMC}_i = \max\left(0.01,\ \text{WMC}_{\text{base}}\left(1 - \lambda \cdot \max\left(H_{\text{norm}}(i),\ A(i)\right)\right)\right). \tag{16}$$

By taking the maximum of the two complementary ambiguity scores, the hybrid reacts to whichever signal—global distribution spread or top-two competition—is dominant for each node, stays in $[0, 1]$, and remains comparable across datasets.

3.5 Comparison of Adaptive Methods

Table 1 summarizes the characteristics of the three proposed strategies.

Table 1 Comparison of ambiguity-aware overlap assignment strategies.

Aspect	**Entropy-Adaptive**	**Margin-Adaptive**	**Hybrid-Adaptive**
Membership information	Full distribution $P(i,:)$	Top-two memberships	Both signals
Ambiguity captured	Global distribution uncertainty	Dominant community competition	Combined
Per-node computation	$O(K)$	$O(K \log K)$	$O(K \log K)$
Suitable scenario	Multi-community ambiguity	Binary community boundary	General purpose

The three strategies capture complementary aspects of membership uncertainty. Entropy is more appropriate when uncertainty is distributed across several communities, margin ambiguity is more targeted toward nodes located between two dominant communities, and the hybrid combines both signals. In our experiments the hybrid provides a robust general-purpose configuration, achieving the best result on DBLP. Algorithm 1 summarizes the complete adaptive overlap assignment procedure.

4 Experiments

This section describes the experimental setup and presents our results. We begin with the datasets (Section 4.1) and baseline configurations (Section 4.2).. We then present the main results comparing adaptive methods against fixed-threshold baselines (Section 4.3), including ablation studies, fairness analysis, computational overhead, statistical significance, and an analysis of the relationship between membership ambiguity and overlap quality.

4.1 Datasets

We evaluate on six benchmark graphs spanning citation networks and heterogeneous networks. Table 2 summarizes their characteristics.

Algorithm 1 Adaptive Overlap Assignment

Require: Soft membership matrix $P \in \mathbb{R}^{N \times K}$, base threshold $\mathrm{WMC}_{\mathrm{base}}$, adaptation strength $\lambda \in [0,1]$, ambiguity type $\tau \in \{\mathrm{Entropy}, \mathrm{Margin}, \mathrm{Hybrid}\}$
Ensure: Adaptive cluster assignments $\mathcal{C}_i$ for all nodes $i = 1, \ldots, N$

for $i \leftarrow 1$ **to** N **do**
  if $\tau = \mathrm{Entropy}$ **then**
    $a_i \leftarrow H_{\mathrm{norm}}(i)$ ▷ Using Eq. (4)
  else if $\tau = \mathrm{Margin}$ **then**
    $a_i \leftarrow 1 - (p_1 - p_2)$ ▷ Using Eq. (5)
  else
    $a_i \leftarrow \max\left(H_{\mathrm{norm}}(i),\ A(i)\right)$ ▷ Using Eq. (15)
  end if
  $\mathrm{WMC}_i \leftarrow \max\left(0.01,\ \mathrm{WMC}_{\mathrm{base}} \times (1 - \lambda a_i)\right)$
  $p_{\max} \leftarrow \max_{c'} P(i, c')$
  $\mathcal{C}_i \leftarrow \{c \mid P(i,c) \geq \mathrm{WMC}_i \times p_{\max}\}$
end for
return $\{\mathcal{C}_i\}_{i=1}^{N}$

Table 2 Dataset characteristics.

Dataset	Nodes	Edges	Classes	Features
Cora	2,708	5,278	7	1,433
CiteSeer	3,327	4,552	6	3,703
PubMed	19,717	44,324	3	500
ACM	2,363	5,318	3	1,870
DBLP	2,591	3,528	4	1,433
IMDB	4,630	54,576	5	1,703

Cora, CiteSeer, and PubMed are citation networks where nodes represent papers and edges represent citation links. ACM, DBLP, and IMDB are bibliographic or entertainment networks (papers–authors–subjects for ACM and DBLP; movies–actors–directors for IMDB) projected into homogeneous graphs.

4.2 Baselines

We compare against three baseline configurations:

1. Cluster-GCN (No Overlap): Crisp clustering with no overlap nodes, representing the lower bound.
2. OCGCN with fixed WMC: The original OCGCN (Amintoosi 2021) with fixed WMC thresholds of 0.10, 0.20, 0.30, 0.40, and 0.50, representing the upper bound when hyperparameters are tuned.
3. Adaptive WMC variants: Entropy-Adaptive, Margin-Adaptive, and their Hybrid combination with $\lambda \in \{0.25, 0.50, 0.75\}$.

To situate the cluster-based results, we additionally evaluate three standard full-batch architectures — GCN, GraphSAGE, and GAT — and a *random-partition* control (the same cluster-based pipeline with random node assignment instead of DANMF). The full-batch models use a comparable architecture to the cluster GCN (16 hidden units per layer, dropout 0.5, Adam lr 0.01) but train on the entire graph with 200 epochs; the random-partition control uses the same per-cluster protocol as the DANMF no-overlap configuration. Table 3 and Figure 1 report micro F1 over 20 seeds.

Table 3 Micro F1 (mean ± std, 20 seeds) of standard full-batch baselines (200 epochs) and the random-partition control, compared with the best cluster-based method per dataset.

Dataset	GCN	GraphSAGE	GAT	Random Part.
Cora	**0.866±0.011**	0.861±0.011	0.865±0.011	0.730±0.016
CiteSeer	**0.727±0.013**	0.723±0.013	0.727±0.011	0.697±0.017
PubMed	0.865±0.004	**0.883±0.004**	0.855±0.004	0.852±0.004
ACM	0.921±0.010	**0.923±0.007**	0.923±0.010	0.876±0.012
DBLP	0.848±0.015	**0.853±0.015**	0.845±0.013	0.807±0.014
IMDB	0.417±0.011	**0.449±0.016**	0.400±0.015	0.374±0.015

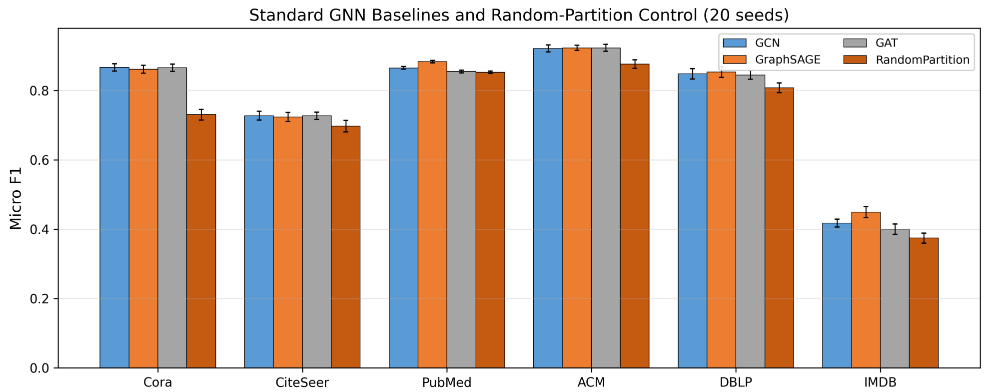


Fig. 1 Standard full-batch GNN baselines and the random-partition control (micro F1, 20 seeds, mean ± std).

Two observations follow. First, full-batch training achieves higher accuracy than cluster-based training on most datasets (e.g., Cora 0.866 vs. 0.830), which is expected: partition-based training trades a modest amount of accuracy for scalability on large graphs, the motivation of Cluster-GCN itself. Second, the random-partition control reveals that clustering quality matters: on Cora, CiteSeer, ACM, DBLP, and IMDB the DANMF no-overlap baseline (0.798, 0.705, 0.910, 0.829, 0.393) exceeds random assignment (0.730, 0.697, 0.876, 0.808, 0.374). PubMed is the exception: DANMF produces a highly imbalanced partition there (the largest cluster contains about 62% of the nodes), and the resulting no-overlap score (0.521) falls below the random-partition control (0.852). This is a limitation of the DANMF clustering on PubMed, not of the overlap selection, and it motivates our adaptive methods, which recover substantial F1 on PubMed (0.628) despite the degraded partition. We discuss this in Section 5.7.

4.3 Results

All experiments use:

- $\mathrm{WMC}_{\mathrm{base}} = 0.3$
- 3-layer GCN with 16 hidden units, ReLU activation, and dropout rate 0.5
- Adam optimizer with learning rate 0.01, 10 epochs per subgraph
- Train/test split: 70%/30% random (per cluster)
- 20 random seeds per configuration for the main comparison and 10 for the ablation study; we report micro F1 (macro F1 is also computed)
- DANMF for community detection with the number of latent communities set to twice the number of classes per dataset (matching the original OCGCN protocol)

The adaptive overlap selection adds negligible computational overhead: entropy and margin computations are $O(K)$ and $O(K \log K)$ per node respectively, applied once after DANMF convergence and before GCN training. Measured wall-clock training times are reported in Table 10 (NVIDIA RTX 3090, mean over 20 seeds).

4.3.1 Adaptive vs. Global WMC

Table 4 presents micro-F1 scores for Cluster-GCN (no overlap), original OCGCN (WMC = 0.30), and the three adaptive methods with $\lambda = 0.50$, averaged over 20 seeds.

Table 4 Micro F1 (mean ± std, 20 seeds) comparison of overlap selection strategies. Best result per dataset in bold.

Dataset	**Cluster-GCN**	**WMC=0.30**	**Entropy (λ=0.5)**	**Margin (λ=0.5)**	**Hybrid (λ=0.5)**
Cora	0.798 ± 0.051	**0.832 ± 0.020**	0.828 ± 0.029	0.830 ± 0.025	0.830 ± 0.030
CiteSeer	0.705 ± 0.053	**0.736 ± 0.030**	0.727 ± 0.044	0.736 ± 0.022	0.729 ± 0.050
PubMed	0.521 ± 0.143	0.612 ± 0.122	0.580 ± 0.136	**0.628 ± 0.127**	0.622 ± 0.133
ACM	**0.910 ± 0.035**	0.879 ± 0.071	0.881 ± 0.080	0.899 ± 0.069	0.895 ± 0.072
DBLP	0.829 ± 0.014	0.844 ± 0.017	0.846 ± 0.024	0.851 ± 0.013	**0.856 ± 0.012**
IMDB	0.393 ± 0.012	0.428 ± 0.015	0.435 ± 0.010	**0.443 ± 0.012**	0.438 ± 0.011
Average	0.693	0.722	0.716	**0.731**	*0.728*

Adaptive methods outperform the no-overlap baseline on all datasets except ACM, where overlap provides no benefit (no-overlap 0.911 vs. best adaptive 0.899). Margin-Adaptive outperforms the untuned default WMC = 0.30 on four of six datasets, with the largest gains on ACM (+2.0 F1), PubMed (+1.6 F1), and IMDB (+1.4 F1).

The Hybrid variant is the best overall method on DBLP (+1.2 F1 over the default). Figure 2 visualizes these results.

Fig. 2 Micro F1 scores across all overlap selection strategies (20 seeds). Bars with red outlines indicate the best result per dataset. Adaptive methods are competitive with or outperform the fixed WMC default on most datasets; on ACM the no-overlap configuration is best.

4.3.2 Ablation: Adaptation Strength

Tables 5, 6 and 7 examine the effect of λ on the three adaptive strategies.

Table 5 Entropy-Adaptive Micro F1 for different λ values (10 seeds). Optimal λ per dataset in bold.

Dataset	λ=0.25	λ=0.50	λ=0.75	Optimal
Cora	0.823	0.822	**0.831**	0.75
CiteSeer	0.735	**0.740**	0.739	0.50
PubMed	0.591	**0.601**	0.582	0.50
ACM	0.864	0.888	**0.902**	0.75
DBLP	0.850	0.843	**0.853**	0.75
IMDB	0.433	0.440	**0.442**	0.75

IMDB, which has the highest mean entropy (0.419) and highest overlap ratio (0.621), benefits from stronger adaptation at $\lambda = 0.75$ across all variants. Margin-Adaptive shows the most per-dataset variation: DBLP prefers the lightest adaptation ($\lambda = 0.25$), while PubMed and IMDB prefer $\lambda = 0.75$. Figures 3, 4 and 5 visualize the lambda sensitivity.

Table 6 Margin-Adaptive Micro F1 for different λ values (10 seeds). Optimal λ per dataset in bold.

Dataset	λ=0.25	λ=0.50	λ=0.75	**Optimal**
Cora	0.817	0.820	**0.825**	0.75
CiteSeer	0.722	**0.745**	0.738	0.50
PubMed	0.604	0.612	**0.645**	0.75
ACM	0.847	**0.907**	0.880	0.50
DBLP	**0.857**	0.851	0.852	0.25
IMDB	0.436	0.441	**0.447**	0.75

Table 7 Hybrid-Adaptive Micro F1 for different λ values (10 seeds). Optimal λ per dataset in bold.

Dataset	λ=0.25	λ=0.50	λ=0.75	**Optimal**
Cora	0.818	**0.825**	0.812	0.50
CiteSeer	0.741	**0.742**	0.738	0.50
PubMed	0.616	0.602	**0.628**	0.75
ACM	0.885	**0.910**	0.870	0.50
DBLP	0.857	**0.863**	0.853	0.50
IMDB	0.433	0.436	**0.448**	0.75

4.3.3 Fair Comparison: Equal Overlap Ratios

A critical question is whether adaptive methods perform better because they select better overlap nodes, or simply because they generate more overlap. Table 8 compares adaptive methods against the closest fixed-WMC baseline at matched overlap ratios.

Table 8 Fair comparison at matched overlap ratios.

Dataset	**Adaptive**	**Overlap**	**Closest Fixed**	**Overlap**	**ΔF1**	**Winner**
Cora	margin	1.44	WMC=0.20	1.48	+0.005	Adaptive
DBLP	hybrid	1.33	WMC=0.20	1.35	+0.013	Adaptive
PubMed	margin	1.31	WMC=0.20	1.35	−0.032	Fixed
ACM	margin	1.31	WMC=0.20	1.35	+0.018	Adaptive
CiteSeer	margin	1.18	WMC=0.30	1.15	+0.000	Tie
IMDB	margin	2.53	WMC=0.20	2.56	+0.003	Adaptive

At equal overlap ratios, adaptive methods win on four of six datasets (CiteSeer is a statistical tie, ΔF1 = +0.000; PubMed is the only loss at −0.032), with gains ranging from +0.003 (IMDB) to +0.018 (ACM). Fixed WMC wins only on PubMed (−0.032), where adaptive methods underperform at matched overlap. This indicates that adaptive selection identifies higher-quality overlap nodes in most cases. Figure 6 visualizes this comparison.

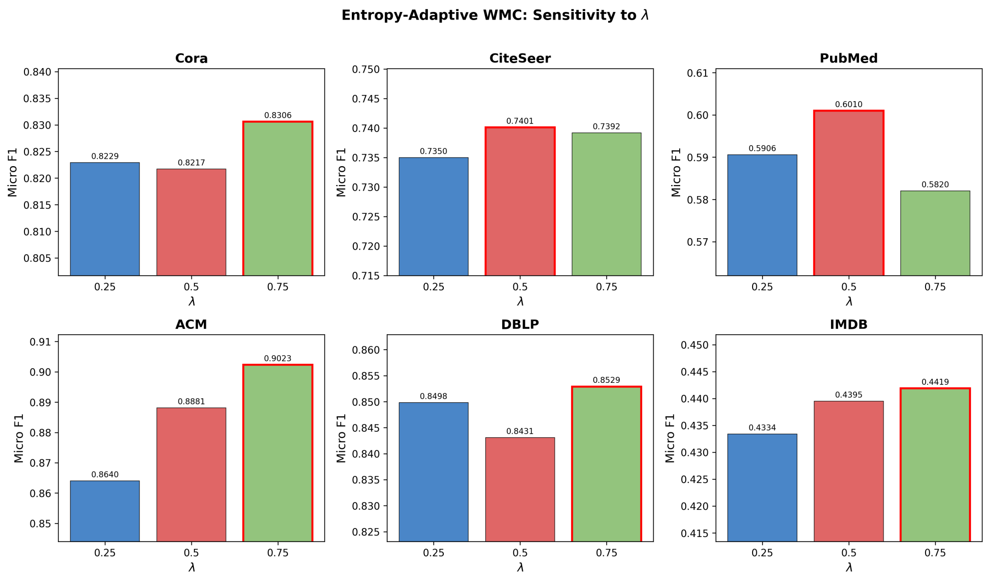


Fig. 3 Entropy-Adaptive WMC performance as a function of λ for each dataset. Red-bordered bars indicate the optimal λ. $\lambda = 0.75$ is optimal for four of six datasets.

4.3.4 Overlap Efficiency

We define overlap efficiency as the F1 gain per unit overlap increase relative to the no-overlap baseline. Table 9 presents this metric for each method.

Table 9 Overlap efficiency (F1 gain / overlap ratio increase) for each method.

Method	Cora	DBLP	PubMed	ACM	CiteSeer	IMDB
WMC=0.10	0.049	0.056	0.252	−0.043	0.092	0.026
WMC=0.20	0.056	0.042	0.402	−0.084	0.122	0.030
WMC=0.30	0.092	0.063	0.342	−0.118	0.204	0.029
Entropy	0.073	0.054	0.190	−0.096	0.128	0.027
Margin	0.073	0.066	0.341	−0.036	0.169	0.032
Hybrid	0.072	0.083	0.317	−0.047	0.129	0.028

Among the fixed thresholds shown, WMC = 0.30 achieves the highest efficiency on Cora (0.092) and CiteSeer (0.204), while WMC = 0.20 is most efficient on PubMed (0.402). Among adaptive methods, Margin-Adaptive is the most efficient on PubMed (0.341) and IMDB (0.032), Hybrid-Adaptive on DBLP (0.083), and Entropy-Adaptive on Cora (0.073, tied with Margin). Efficiency is negative for every method on ACM, consistent with overlap providing no benefit on that dataset. Averaged across all six datasets, Margin-Adaptive achieves the highest mean efficiency (0.108), edging out the best fixed threshold WMC = 0.30 (0.102), which indicates that adaptive methods

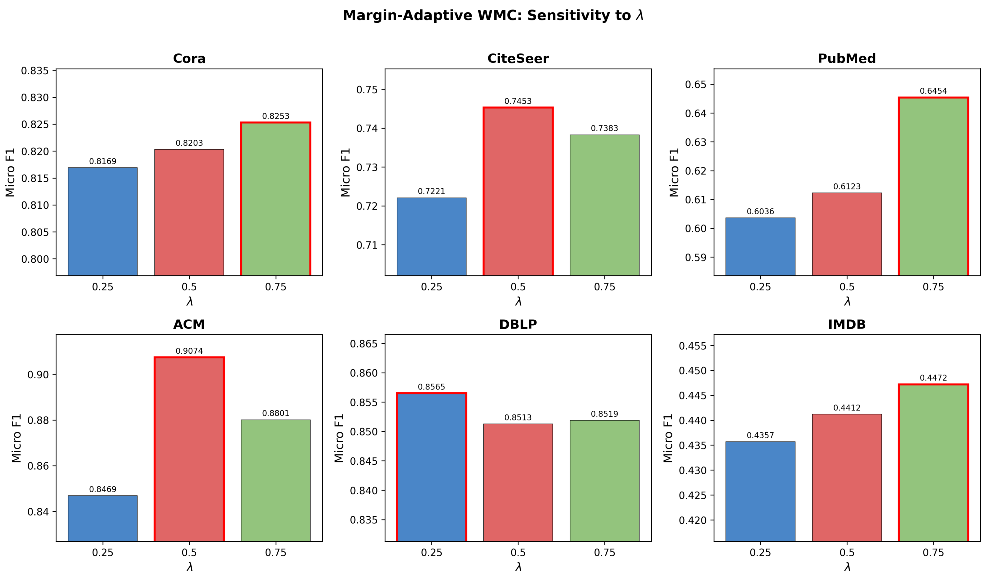


Fig. 4 Margin-Adaptive WMC performance as a function of λ for each dataset. Red-bordered bars indicate the optimal λ. Margin-Adaptive shows the most per-dataset variation.

extract slightly more F1 gain per unit of overlap added. Figure 7 visualizes overlap efficiency across methods.

4.3.5 Training Time and Computational Overhead

Table 10 reports the measured wall-clock training time per run (mean over 20 seeds) for the main configurations, together with the one-time DANMF decomposition cost. The adaptive selection itself adds negligible overhead: entropy and margin scores are computed once per node in $O(K)$ and $O(K \log K)$, adding at most a few milliseconds to the pipeline. Training time is dominated by the GCN epochs and grows with the size of the cluster subgraphs; because adaptive methods produce slightly larger subgraphs (more overlap nodes), their per-run time is marginally higher than the no-overlap configuration (e.g., Cora: 1.59 s no-overlap vs. 1.62 s margin-adaptive), but remains well below the one-time DANMF cost on the larger graphs.

Figure 8 visualizes the per-run training time and the one-time decomposition cost.

4.3.6 Statistical Significance

Paired t-tests and Wilcoxon signed-rank tests comparing adaptive methods against the untuned default WMC $= 0.30$ (20 seeds each), together with pooled tests across all 120 paired runs, are reported in Table 11.

Against the untuned default WMC $= 0.30$, Hybrid-Adaptive is significantly better on DBLP ($p = 0.020$) and Margin-Adaptive on IMDB ($p = 0.003$). Aggregating the six per-dataset mean differences, the gain of Margin-Adaptive over the default is positive

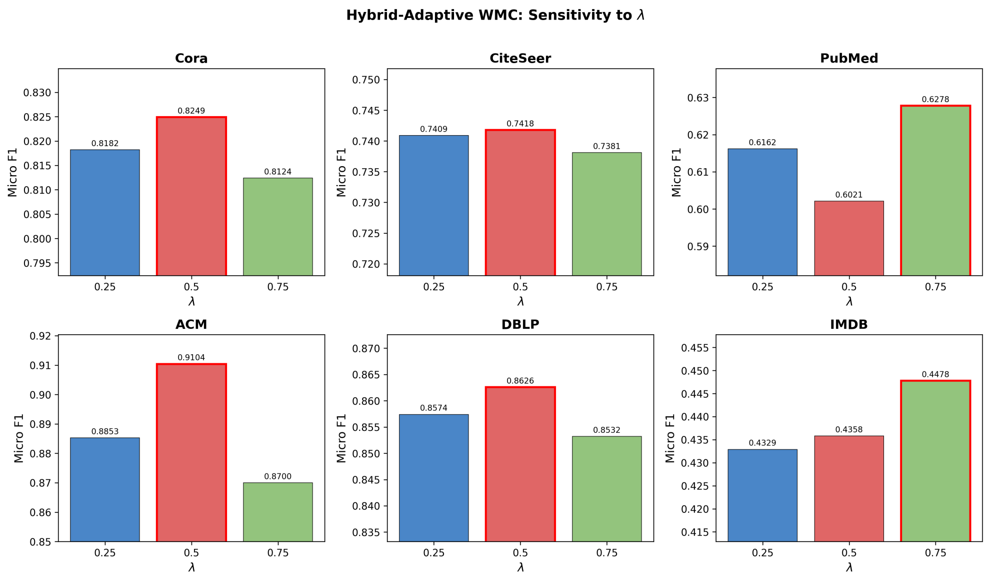


Fig. 5 Hybrid-Adaptive WMC performance as a function of λ for each dataset. Red-bordered bars indicate the optimal λ. $\lambda = 0.50$ is optimal for four of six datasets, making the hybrid the most stable configuration.

Table 10 Training time (seconds, mean over 20 seeds; DANMF fit is a single one-time measurement) on an NVIDIA RTX 3090.

Dataset	DANMF fit	No overlap	WMC=0.30	Margin	Hybrid	Entropy
Cora	2.5	1.59	1.60	1.62	1.68	1.63
CiteSeer	2.5	1.56	1.61	1.67	1.72	1.67
PubMed	36.1	1.41	1.87	2.00	2.33	2.19
ACM	2.4	0.80	0.91	0.86	1.04	0.97
DBLP	1.7	1.05	0.98	1.01	1.13	1.09
IMDB	22.1	1.62	2.42	2.53	2.52	2.48

but not significant at the 0.05 level ($t = 2.40$, $p = 0.061$). Pooled across all 120 paired runs (20 seeds $\times$ 6 datasets), it is statistically significantly better than the default (paired t-test $t = 2.32$, $p = 0.022$; Wilcoxon $p = 0.020$).

This statistical picture supports the paper's core framing: adaptive thresholding does not universally beat oracle-tuned baselines, but it is significantly better than a fixed default threshold and eliminates the need for WMC tuning. Figure 9 provides a comprehensive view of the micro-F1 difference between margin-adaptive and every fixed WMC baseline.

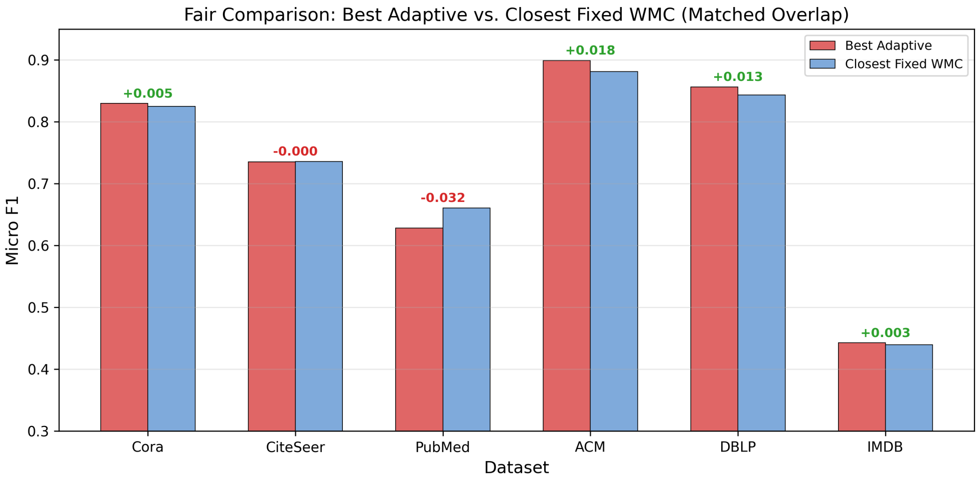


Fig. 6 Fair comparison at matched overlap ratios (micro F1). Green annotations indicate adaptive wins; red indicates fixed wins. Adaptive methods win on four of six datasets (CiteSeer is a statistical tie and PubMed the only loss), with the largest gain on ACM (+0.018).

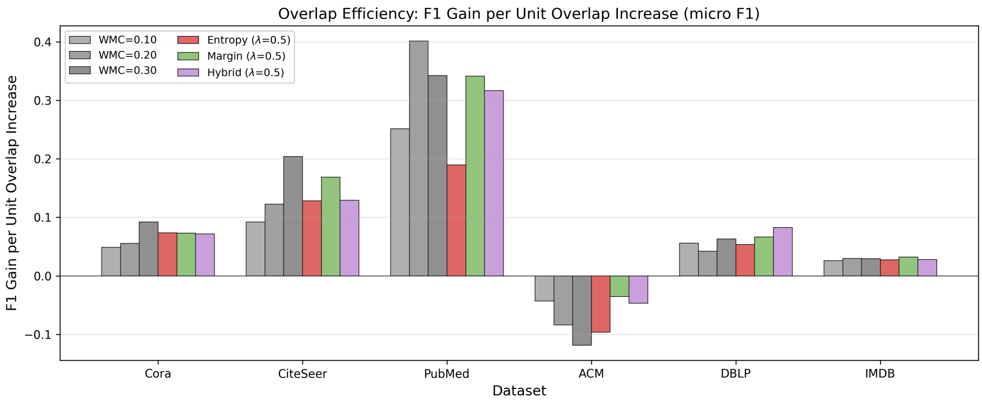


Fig. 7 Overlap efficiency (F1 gain per unit overlap increase, micro F1) for each method. Adaptive methods match fixed thresholds in efficiency on most datasets; Margin-Adaptive is the most efficient adaptive variant on PubMed and IMDB.

4.3.7 Ambiguity Analysis

Dataset Ambiguity Statistics.

Table 12 presents the membership ambiguity statistics derived from DANMF output across all datasets.

IMDB has the highest entropy ($\mu = 0.419$) and overlap ratio (0.621), reflecting its dense structure and role-heterogeneous nature. CiteSeer has the lowest entropy

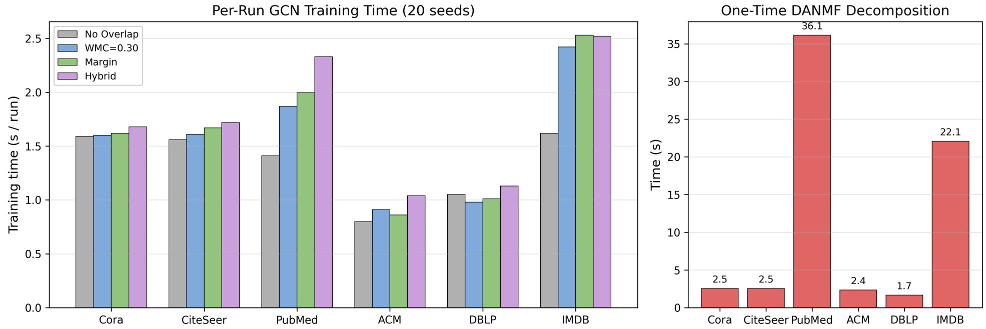


Fig. 8 Left: mean GCN training time per run (20 seeds) for no-overlap, WMC=0.30, margin-adaptive, and hybrid-adaptive. Right: one-time DANMF decomposition cost per dataset. Training time is dominated by the GCN and grows with subgraph size; the adaptive selection itself adds negligible overhead.

Table 11 Paired statistical tests (micro F1, 20 seeds): best adaptive method vs. the untuned default WMC = 0.30, plus pooled tests across all 120 paired runs.

Setting	Adaptive	vs	t-stat	p-val	Wilcoxon p	Winner
DBLP	hybrid	WMC=0.30	2.538	0.020	0.036	Adaptive
IMDB	margin	WMC=0.30	3.402	0.003	0.003	Adaptive
IMDB	hybrid	WMC=0.30	2.285	0.034	0.048	Adaptive
PubMed	margin	WMC=0.30	1.025	0.318	0.452	No diff
ACM	margin	WMC=0.30	1.428	0.169	0.189	No diff
Cora	margin	WMC=0.30	−0.347	0.733	0.784	No diff
CiteSeer	margin	WMC=0.30	−0.074	0.942	1.000	No diff
Pooled (120)	margin	WMC=0.30	2.322	0.022	0.020	Adaptive
Pooled (120)	hybrid	WMC=0.30	1.794	0.075	0.041	Adaptive (Wilcoxon)

($\mu = 0.071$), indicating near-deterministic memberships. Entropy variance is highest on IMDB ($\sigma = 0.280$) and lowest on CiteSeer ($\sigma = 0.126$), suggesting that IMDB has the widest spread of ambiguity levels—the condition under which adaptive thresholding should have the most room to help.

Correlation Between Ambiguity and Overlap.

Table 13 reports Spearman rank correlations between ambiguity measures and structural properties. All correlations are significant at $p < 0.001$.

Entropy correlates strongly with both overlap assignment and cluster count across all datasets ($r = 0.60$–0.81), confirming that high-entropy nodes are systematically assigned to more clusters. Ambiguity correlates strongly with overlap on four of six datasets but weakly on CiteSeer and DBLP ($r \approx 0.10$), where entropy and margin measure fundamentally different aspects of the membership distribution. On PubMed, entropy and ambiguity are nearly identical ($r = 0.981$), suggesting the two measures are redundant on that dataset.

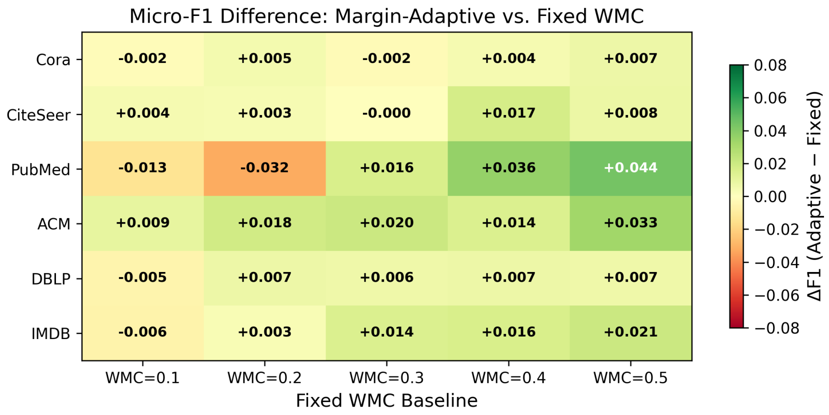


Fig. 9 Micro-F1 difference between Margin-Adaptive and each fixed WMC baseline. Green cells indicate adaptive wins; red cells indicate fixed wins. Margin-Adaptive is competitive with most fixed WMC values across datasets without tuning.

Table 12 Membership ambiguity statistics per dataset (DANMF output, WMC = 0.30 overlap).

Dataset	Entropy μ	Entropy σ	Ambiguity μ	Overlap Ratio	Mean Clusters/Node
Cora	0.138	0.167	0.305	0.279	1.37
CiteSeer	0.071	0.126	0.467	0.137	1.15
PubMed	0.143	0.166	0.187	0.230	1.27
ACM	0.159	0.190	0.256	0.240	1.26
DBLP	0.122	0.173	0.522	0.211	1.24
IMDB	0.419	0.280	0.500	0.621	2.20

The scatter plots in Figure 10 visualize the entropy–cluster count relationship across all datasets, showing the clear positive trend.

Stratified Ambiguity Analysis.

To directly test the ambiguity hypothesis, we stratify nodes into entropy terciles and examine their overlap assignment rates. Table 14 presents the results.

The pattern is consistent across all datasets: high-entropy nodes receive 41–95% overlap assignment, while low-entropy nodes receive 0–17%. On IMDB, the most ambiguous nodes are assigned to 3.4 clusters on average, confirming they genuinely straddle multiple communities. On CiteSeer and DBLP, where many nodes have near-zero entropy, the two lowest terciles merge into a single low-entropy bin with almost no overlap. Figure 11 visualizes this relationship, showing the stark contrast between entropy terciles.

Why Adaptive WMC Helps Some Datasets More Than Others.

Table 15 examines the relationship between entropy statistics and adaptive performance gain.

Table 13 Spearman rank correlations between ambiguity measures and structural properties.

Dataset	Entropy↔Overlap	Entropy↔Clusters	Ambiguity↔Overlap	Entropy↔Ambiguity
Cora	0.706	0.716	0.600	0.528
CiteSeer	0.601	0.602	0.097	−0.366
PubMed	0.693	0.697	0.729	0.981
ACM	0.696	0.698	0.613	0.624
DBLP	0.677	0.680	0.099	−0.398
IMDB	0.701	0.808	0.790	0.676

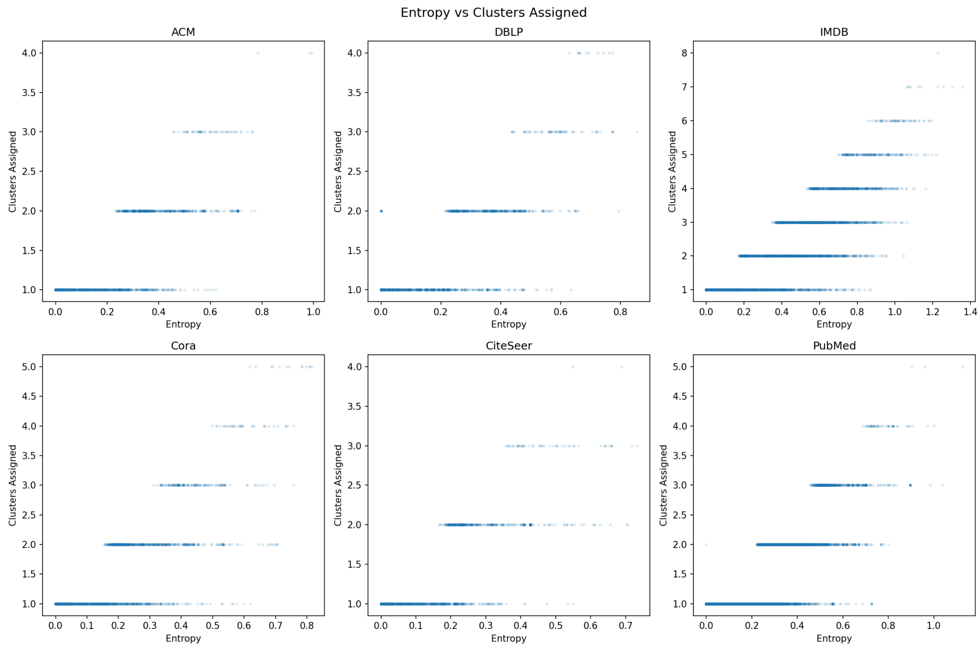


Fig. 10 Entropy vs. number of clusters assigned for each dataset. Each point represents a node. Higher entropy correlates with more cluster assignments, confirming that entropy captures genuine membership ambiguity.

Performance gains from adaptive thresholding show a partial relationship with entropy variance: the largest gains occur on datasets with higher variance (ACM, PubMed, IMDB), while CiteSeer, with the lowest variance, gains nothing. The relationship is not strictly monotonic—variance alone does not determine the gain, because the benefit also depends on how far the default WMC = 0.30 is from the per-dataset optimum. For example, Cora (variance 0.028) shows negligible gain because WMC = 0.30 is already the best fixed threshold (0.832 micro F1), leaving no room for improvement regardless of method. PubMed and DBLP have similar entropy variance to

Table 14 Overlap ratio and mean clusters per entropy tercile.

Dataset	Tercile	Entropy Range	Nodes	Overlap Ratio	Mean Clusters
Cora	Low	0–0.0001	903	0.000	1.00
	Medium	0.0001–0.19	902	0.076	1.08
	High	0.19–0.81	903	0.761	2.04
CiteSeer	Low+Med	0–0.03	2218	0.000	1.00
	High	0.03–0.73	1109	0.411	1.46
PubMed	Low	0–0.009	6570	0.000	1.00
	Medium	0.009–0.20	6575	0.000	1.00
	High	0.20–1.13	6572	0.689	1.80
ACM	Low	0–0.005	788	0.000	1.00
	Medium	0.005–0.23	787	0.000	1.00
	High	0.23–0.99	788	0.718	1.79
DBLP	Low+Med	0–0.15	1727	0.006	1.01
	High	0.15–0.86	864	0.620	1.72
IMDB	Low	0–0.26	1544	0.170	1.17
	Medium	0.26–0.56	1542	0.739	2.02
	High	0.56–1.36	1544	0.953	3.42

Table 15 Relationship between entropy statistics and adaptive performance gain.

Dataset	Entropy Variance	Gain vs. Default (F1 pts)	Interpretation
ACM	0.036	+2.0	Many ambiguous nodes; largest absolute gain
PubMed	0.027	+1.6	High variance → many ambiguous nodes → large gain
IMDB	0.079	+1.4	Highest mean + variance → largest overlap ratios
DBLP	0.030	+1.2	Moderate variance → modest gain
Cora	0.028	−0.2	Near-optimal default → no room regardless of method
CiteSeer	0.016	+0.0	Very low entropy → minimal room for improvement

Cora ($\sigma^2 \approx 0.03$), yet gain +1.6 and +1.2 F1, respectively, because their default WMC = 0.30 underperforms the per-dataset optimum (WMC = 0.20 for PubMed, WMC = 0.10 for DBLP), and adaptive methods can lower the threshold for ambiguous nodes to recover the lost performance. The dashed trend line in Figure 12 shows the linear fit between entropy variance and performance gain, illustrating the overall positive association despite the outliers. Figure 12 visualizes this relationship.

Ceiling effects (ACM), near-optimal defaults (Cora), and very low ambiguity (CiteSeer) limit gains regardless of variance.

5 Discussion

This section interprets the experimental findings, discusses their implications, and outlines limitations and future directions. We first validate the ambiguity hypothesis and

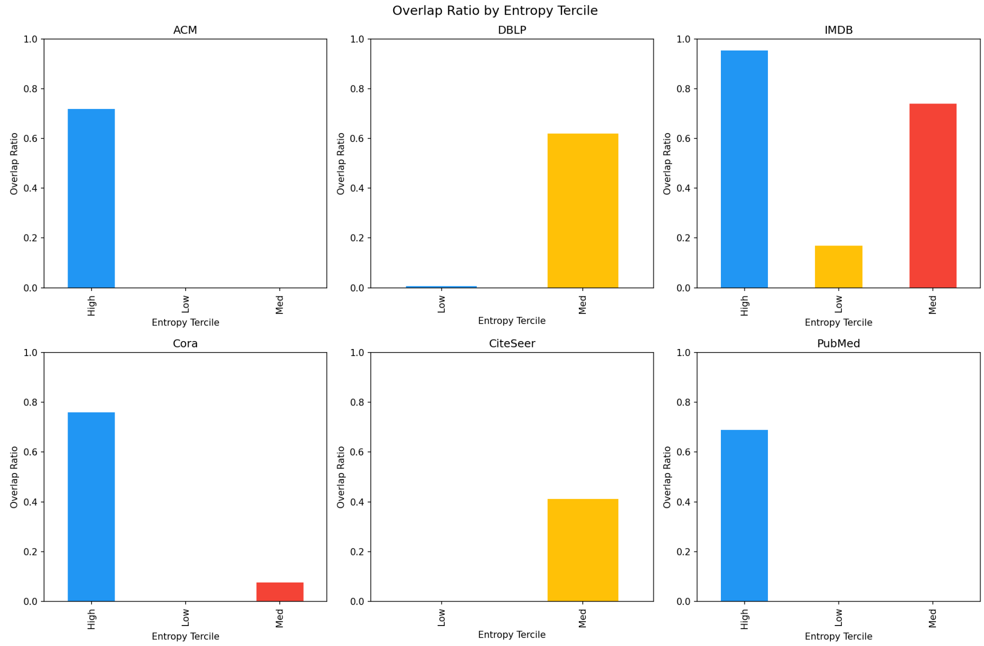


Fig. 11 Overlap ratio by entropy tercile for each dataset. High-entropy nodes consistently receive far more overlap assignments than low-entropy nodes, validating the hypothesis that membership ambiguity identifies useful overlap candidates.

the practical value of eliminating the WMC hyperparameter (Sections 5-5.5), followed by practical recommendations (Section **??**) and threats to validity (Section 5.7).

5.1 The Ambiguity Hypothesis Is Validated

The stratified analysis and correlation studies provide strong evidence that community-membership ambiguity identifies useful overlap nodes. High-entropy nodes are 41–95% overlap candidates, and they are assigned to 1.5–3.4 clusters on average. The mechanism is intuitive: nodes with ambiguous memberships genuinely participate in multiple communities, and including them in multiple cluster subgraphs provides the GCN with richer training signals. Low-entropy nodes, by contrast, are well-served by single-cluster training.

5.2 Adaptive Thresholding Eliminates a Hyperparameter

The primary practical contribution is the elimination of the WMC hyperparameter search. Our fixed-WMC comparison (Table 4) shows that the optimal WMC varies across datasets: WMC = 0.30 is best for Cora and CiteSeer, WMC = 0.20 for PubMed, and WMC = 0.10 for DBLP and IMDB (and no overlap at all on ACM). Without adaptive methods, practitioners must search over WMC values—a costly process that requires retraining on each dataset. Margin-Adaptive (or the Hybrid)

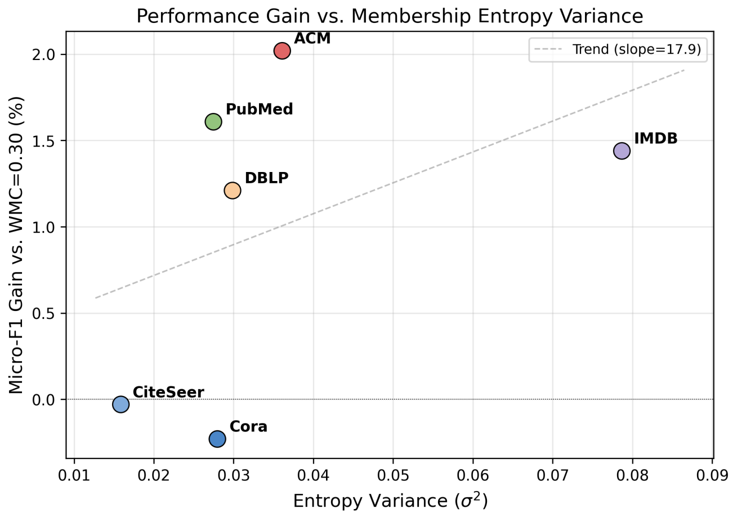


Fig. 12 Performance gain (micro-F1 improvement over WMC = 0.30) vs. entropy variance (σ^2) for each dataset. Datasets with higher entropy variance (ACM, PubMed, IMDB) show the largest gains, while CiteSeer, with the lowest variance, gains nothing.

with $\lambda = 0.50$ achieves competitive performance across all six datasets without per-dataset tuning, and is significantly better than the fixed default in the pooled paired analysis ($p = 0.022$). Importantly, this is not a claim of universal superiority over well-tuned baselines, but rather a demonstration that a single universal configuration ($\text{WMC}_{\text{base}} = 0.3$, $\lambda = 0.5$) performs comparably to per-dataset tuned WMC, eliminating the need for grid search. This is particularly valuable in scenarios where practitioners work with many graphs or where the optimal WMC is not known a priori. Furthermore, while λ is technically a hyperparameter, it acts as a global robustness knob rather than a dataset-specific threshold. As shown in the ablation study (Tables 5, 6, and 7), setting $\lambda = 0.50$ consistently yields near-optimal or optimal results across all six diverse datasets without any per-dataset tuning. This contrasts sharply with the fixed WMC baseline, where the optimal value varies drastically across datasets (e.g., 0.10 for DBLP/IMDB vs. 0.30 for Cora/CiteSeer). The sensitivity to λ is substantially lower than the sensitivity to WMC, thus practically eliminating the need for a grid search.

5.3 Performance Gains Are Modest but Real

At matched overlap ratios, adaptive methods improve by +0.003 to +0.018 F1 over fixed baselines on four of six datasets.

The gains are real but modest, reflecting the fact that the original WMC already captures a substantial portion of the useful overlap signal. Adaptive methods refine this signal by scaling the threshold based on node-level ambiguity, but the marginal improvement is bounded. The contribution is not universal performance superiority over well-tuned baselines, but rather principled threshold selection that eliminates the tuning cost.

5.4 Entropy vs. Margin: Complementary Signals

Entropy-Adaptive, Margin-Adaptive, and the Hybrid variant perform similarly overall, but show different per-dataset preferences. On datasets where ambiguity is primarily a top-two competition, Margin-Adaptive performs best with light adaptation on DBLP ($\lambda = 0.25$) and moderate adaptation on CiteSeer ($\lambda = 0.50$). On datasets with richer multi-cluster ambiguity (PubMed, IMDB), stronger adaptation ($\lambda = 0.75$) helps all variants. The weak correlation between entropy and ambiguity on DBLP and CiteSeer ($r \approx -0.4$) confirms that these measures capture genuinely different aspects of the membership distribution. The Hybrid variant combines both signals and achieves the best overall result on DBLP, confirming their complementarity.

5.5 Community-Detection Agnosticism

While DANMF was utilized in this study to generate the soft membership matrix P, the proposed adaptive thresholding mechanism is fundamentally agnostic to the underlying community detection algorithm. As long as the chosen clustering method outputs a normalized membership matrix $P \in \mathbb{R}^{N \times K}$ whose rows sum to one, the adaptive WMC formulas (Entropy, Margin, and Hybrid) can be applied directly without any modification. This makes Adaptive-OCGCN a general-purpose plugin that can improve partition-based GNN training pipelines utilizing various soft clustering techniques.

5.6 Practical Recommendations

Based on our findings, we recommend:

1. Default configuration: Margin-Adaptive (or the Hybrid) with $\mathrm{WMC}_{\mathrm{base}} = 0.3$ and $\lambda = 0.50$. This single configuration is significantly better than the fixed default in the pooled paired analysis ($p = 0.022$), outperforms it on four of six datasets, and provides competitive performance across all tested datasets without per-dataset tuning.
2. When strong ambiguity is expected: Increase λ to 0.75 (as on IMDB).
3. When a tuned fixed WMC is available: Adaptive methods are unlikely to significantly outperform it, but they provide a principled starting point that avoids the tuning cost.

5.7 Threats to Validity

Dataset scope.

We evaluate on six datasets, all of moderate size (2,363–19,717 nodes). The largest graph, PubMed, has 19,717 nodes, which is relatively small by modern GNN standards, even though the paper title emphasizes scalable GNNs. It is crucial, however, to distinguish between the scalability of the proposed adaptive thresholding mechanism and the scalability of the upstream clustering algorithm. The adaptive WMC framework itself is theoretically highly scalable: the node-level ambiguity scores add only $O(NK)$ or $O(NK \log K)$ computational overhead, which is negligible compared with GCN training. The current evaluation is limited to standard benchmark sizes because the upstream dependency—specifically the DANMF clustering algorithm used to generate P—becomes a computational bottleneck on graphs with millions of nodes. Therefore, the proposed method is fully prepared for large-scale graphs, provided it is paired with a more scalable soft community detection algorithm in future work. Scaling to larger graphs (e.g., OGB benchmarks with millions of nodes) would further strengthen the claims and test whether the ambiguity signal remains informative at scale.

Community detection method.

All experiments use DANMF for membership estimation. Different community detection methods may produce different membership distributions, potentially affecting the ambiguity signal. The adaptive framework is method-agnostic in principle, but the specific entropy and margin values depend on the quality of the membership estimation. On PubMed, DANMF yields a highly imbalanced partition (the largest cluster contains about 62% of the nodes), which degrades the cluster-based no-overlap baseline (0.521 micro F1) below a random-partition control (0.852); the adaptive overlap methods still recover substantial accuracy (0.628) in this regime. This is a limitation of the clustering step rather than of overlap selection, and it suggests that balancing cluster sizes (e.g., via size constraints or a different K) is a promising avenue for future work.

Overlap ratio range.

Our experiments fix $\mathrm{WMC}_{\mathrm{base}} = 0.3$ and vary only λ. Different base WMC values could shift the overlap ratio range and affect the relative advantage of adaptive methods. We chose $\mathrm{WMC}_{\mathrm{base}} = 0.3$ because it is the default in the original OCGCN.

Statistical power.

With 20 seeds per configuration in the main comparison (10 in the ablation), we detected a significant pooled improvement of Margin-Adaptive over the default threshold ($p = 0.022$). Non-significant per-dataset differences on Cora, PubMed, and ACM could still reflect limited power to detect small effect sizes; larger-scale experiments with more seeds would clarify this.

Single GCN architecture.

We use a fixed GCN with 16 hidden units per layer. Different architectures (deeper/wider GCNs, attention-based models) might interact differently with overlap assignment. The adaptive thresholding is architecture-agnostic, but the magnitude of its benefit may vary.

5.8 Conclusion

This paper introduces adaptive overlap thresholding for Overlapping Cluster-GCN, replacing the global WMC threshold with node-specific thresholds derived from community-membership ambiguity. Three variants are proposed: Entropy-Adaptive

WMC, which scales the threshold by normalized membership entropy, Margin-Adaptive WMC, which scales by the ambiguity of the top-two membership gap, and Hybrid-Adaptive WMC, which combines both signals. All three replace per-dataset WMC grid search with a single universal configuration ($\mathrm{WMC}_{\mathrm{base}} = 0.3$, $\lambda = 0.5$) that achieves competitive performance across six benchmark graphs; Margin-Adaptive is significantly better than the fixed default in the pooled paired analysis ($p = 0.022$).

The core hypothesis—that membership ambiguity identifies useful overlap nodes—is validated through stratified analysis showing that high-entropy nodes are 41–95% overlap candidates, with strong correlations between entropy and cluster count on all datasets. Performance gains from adaptive thresholding are modest but real: at matched overlap ratios, adaptive methods improve by +0.003 to +0.018 F1 over fixed baselines on four of six datasets, and Margin-Adaptive is significantly better than the fixed default in the pooled paired analysis ($p = 0.022$).

The practical value of adaptive thresholding is the removal of the WMC hyperparameter search, a process that varies across datasets and requires costly retraining. Margin-Adaptive (or the Hybrid) with $\lambda = 0.50$ provides a robust default that works well across diverse graph types. Future work should explore scaling to larger graphs, per-node λ adaptation, and theoretical analysis of the overlap quality bound.

References

Amintoosi, M.: Overlapping clusters in cluster graph convolutional networks. Journal of Algorithms and Computation **53**(2), 33–45 (2021)

Chiang, W.-L., Liu, X., Si, S., Li, Y., Bengio, S., Hsieh, C.-J.: Cluster-gcn: An efficient algorithm for training deep and large graph convolutional networks. In: Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, pp. 257–266 (2019)

Chen, J., Ma, T., Xiao, C.: Fastgcn: Fast learning with graph convolutional networks via importance sampling. In: Proceedings of the International Conference on Learning Representations (ICLR) (2018)

Dai, H., Dai, B., Song, L.: Stochastic training of graph convolutional networks with variance reduction. In: Proceedings of the 35th International Conference on Machine Learning (ICML), pp. 942–950 (2018)

Gawlikowski, J.e.a.: A survey of uncertainty in deep neural networks. Artificial Intelligence Review (2023) https://doi.org/10.1007/s10462-023-10562-9

Guang, M.e.a.: Graph convolutional networks with adaptive neighborhood awareness. IEEE Transactions on Pattern Analysis and Machine Intelligence (2024) https://doi.org/10.1109/TPAMI.2024.3391356

He, C.e.a.: Semi-supervised overlapping community detection in attributed graph with graph convolutional autoencoder. Information Sciences (2022) https://doi.org/10.

1016/j.ins.2022.07.036

Hamilton, W.L., Ying, R., Leskovec, J.: Inductive representation learning on large graphs. In: Advances in Neural Information Processing Systems (NeurIPS), pp. 1024–1034 (2017)

Kipf, T.N., Welling, M.: Semi-supervised classification with graph convolutional networks. In: Proceedings of the International Conference on Learning Representations (ICLR) (2017)

Kwon, Y.e.a.: Uncertainty-aware prediction of chemical reaction yields with graph neural networks. Journal of Cheminformatics (2022) https://doi.org/10.1186/s13321-021-00579-z

Li, X.e.a.: Dual graph neural network for overlapping community detection. The Journal of Supercomputing (2023) https://doi.org/10.1007/s11227-023-05435-5

Moradan, A.e.a.: Ucode: unified community detection with graph convolutional networks. Machine Learning (2023) https://doi.org/10.1007/s10994-023-06402-0

Park, S.e.a.: Confidence-aware graph neural networks for learning reliability assessment commitments. IEEE Transactions on Power Systems (2023) https://doi.org/10.1109/TPWRS.2023.3298735

Peng, H.e.a.: Reinforced neighborhood selection guided multi-relational graph neural networks. ACM Transactions on Information Systems (2021) https://doi.org/10.1145/3490181

Sismanis, K.e.a.: Overlapping community detection using graph attention networks. Future Generation Computer Systems (2024) https://doi.org/10.1016/j.future.2024.107529

Shang, Y., Zhao, L., Xu, J., Gao, L.: Overlapping community detection based on modularity and labeled propagation. Physics Procedia **57**, 75–83 (2014)

Tran, Q.M.A., Duong, H., Phung, D.: Danmf: Deep adversarial non-negative matrix factorization for community detection. arXiv preprint arXiv:1810.11496 (2018)

Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., Bengio, Y.: Graph attention networks. In: Proceedings of the International Conference on Learning Representations (ICLR) (2018)

Yuan, S.e.a.: Overlapping community detection on complex networks with graph convolutional networks. Computer Communications (2022) https://doi.org/10.1016/j.comcom.2022.12.008

Statements & Declarations

Funding

The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Competing Interests

The authors have no relevant financial or non-financial interests to disclose.